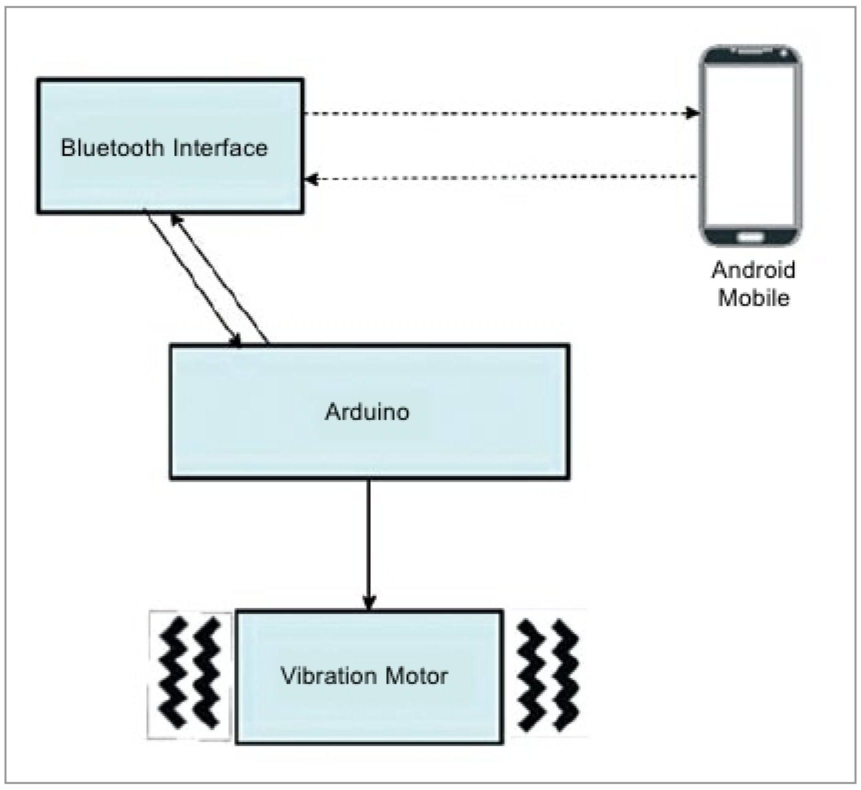


Fig. 2: Notification message

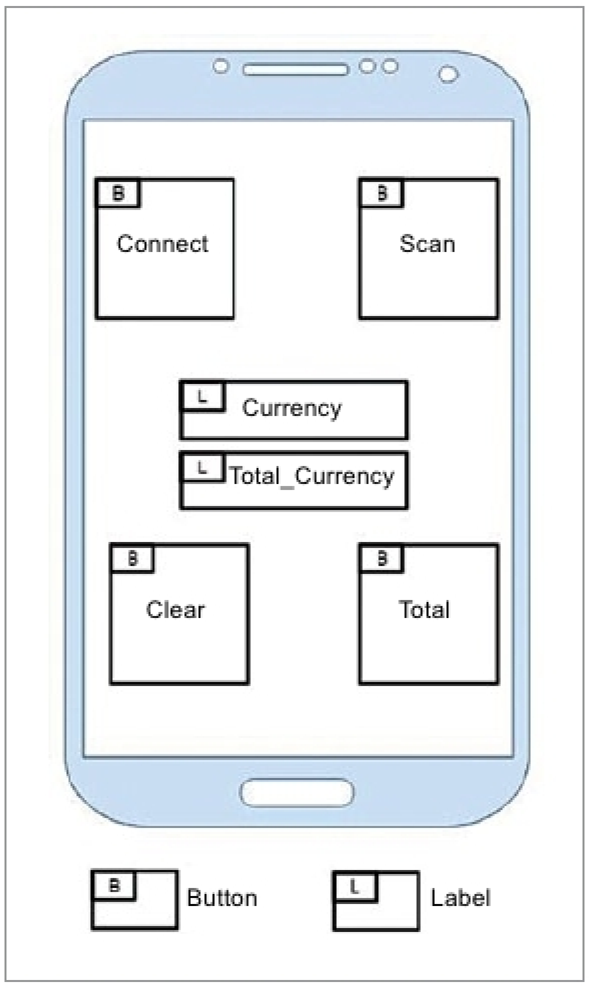


Fig. 3: Mobile app

is complete, the device vibrates in a specific sequence and the smartphone connected to the device via the Bluetooth interface gives the sum as voice output to the user.

The vibration sequence is unique for every different denomination of the currency. The device can be concealed as a wallet never giving off its purpose. This guarantees anonymity and a sense of privacy and security to the user.

The embedded system known as the smart vision wallet has been solely created with the desire to assist the blind community to live alongside others as equals. The device has been fashioned in the shape of a common wallet, and weighs about the same. It works by placing the currency notes inside, after pressing a button on the wallet. The components within the wallet scan the currency for their denomination. These also check whether the currency is original or counterfeit.

Output is given via vibrations—a vibration pattern is assigned for each currency denomination—or through audio output. The wallet connects to Android devices via Bluetooth. An Android app has also been developed that works with the device. It helps in scanning the currency or finding the total currency that has been previously scanned.

This system does not require much effort by the user, who only has to place the currency and press the button on the device or the Android app.

Such a device gives the visually-impaired independence in finding the money. It eliminates the need for another person's help in finding the correct denomination. Information regarding the currency is provided in a discreet manner. Due to the device's portability feature and the ability to differentiate between real and counterfeit money, it is useful for everyone. It could also incorporate existing technologies used in smart wallets that are currently available in the market.

## Features of smart vision wallet

Smart vision wallet has the following features:

- Currency differentiation
- Counterfeit detection
- Notification

***Currency differentiation.*** A smart vision wallet must meet two requirements: affordability and portability. The device should also look like an ordinary wallet. Primary difference between the different denominations of Indian currency is colour. Hence, the most effective and simple means to identify the denominations of Indian currency is the ability to detect these unique differences in colour, and to do that, colour sensors having great precision are required.

***Counterfeit detection.*** To check the validity of security features implemented in the Indian currency, the smart vision wallet makes use of colour sensors and programmed ATmega328P MCU.

***Notifications.*** The user needs to be informed of the counted sum or the detected counterfeit. The smart vision wallet is designed to be used by the visually-impaired and the hearing-impaired community. There are two ways in which users can be notified: announcement by a pattern of tones and generation of sequences of vibration. The sequence is different for different denominations.

Voice output is designed for the visually-impaired, whereas vibration output is designed for the visually- and hearing-impaired, in addition to being a privacy measure. It is generated using an Android app specifically for the visually-impaired.

The user interface of the app used to generate audio output is designed in such a way that it can be used by the blind with great ease. The app consists of four buttons located at four corners of the screen. This design guarantees ease of use by the visually-impaired. The four buttons are Connect, Scan, Total and Clear.

The Android app consists of four modules: connection, scanning, computing and resetting. Connection module handles the connection of a smartphone with Arduino. Scanning module sends the scan command to smart vision wallet, retrieves denomination and announces it as voice output. EFY